Name : AKASH SAMPATH

Title : Research_Q_A_PART_1_REST_Concept

Q8) . What is REST standard or recommendation for Web applications?

- REST is know as Representational State Transfer
- It is a set of best practices and design recommendations for building we APIs or web services so that different systems can communicate easily over HTTP
- This is not strict rulebook but a recommended standard style used by most modern web applications and APIs
- REST tell develops how to design your API URLs , methods and responses in clean , predictable and scalable way

Runs and Uses normal HTTP methods :

- GET
- POST
- PUT
- DELETE

REST Principals :

Use Resources like Nouns In URL and not verbs:

- /users
- /orders
- /products

Not recommended :

- /getUsers
- /createOrder
- deleteProduct

Note :

- Think URL is resource name and not action name

REST Recommends using HTTPS method to perform actions:

Method / Purpose / Example :

- GET used to read data and get user info
- POST is used for creating new data by adding new user
- PUT used for updating full data by updating user
- PATCH uses for updating partial data like changing email only
- DELETE uses to remove data by deleting user

REST using Stateless Communication :

- REST recommends that each request should be independent
- If server does not remember previous requests
- Every request contains all required information
- No need session memory

REST using Proper HTTP Status Codes :

- REST recommends returning correct response codes

Code : and Meaning :

200 : Means success

201: Means created

400: Means bad request

401: Meaning unauthorized

404: Meaning not found

500: Meaning server error

REST using JSON format for Responses:

- It is a common practice and REST recommends using JSON because of it is simple and lightweight

Example : Code: Response : Using JSON :

```
{
 “id”: 1,
 “name”: “Akash”,
 “email”: “akash@example.com”
}
```

Full Simple Example : REST API for Users :

Resource : Users

- **Get all the users**

Example : Code :

```
GET /users
```

Response: Code : Using JSON :

```
[
  {"id":1,"name":"Akash"},
  {"id":2,"name":"Ravi"}
]
```

Get one user:

Example : code :

```
GET /users/1
```

Create new user: Using JSON : Code : Body :

```
{
  "name": "John"
}
```

Update user:

Example : Code :

```
PUT /users/1
```

Delete user:

Example : Code :

DELETE /users/1

Summary :

```
GET     /users        → get all users
GET     /users/1      → get one user
POST    /users        → create user
PUT     /users/1      → update user
PATCH   /users/1      → partial update
DELETE  /users/1      → delete user
```

WHY REST is Recommended for Web Applications :

- It is easy to understand by using standard HTTP which is scalable and has language independent and works well with all browsers easily where this is widely supported and has clean structure